Cantera vs NN | Test IC 88 | NRMSE (util.py)

T0 = 1502.1 K P0 = 4.980 atm phi0 = 0.859 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0536	0.764	5.02	0.609	0.392	0.914	0.0409	1.27	4.48	0	Complete
2 P-only DeepONet10	**10k**	0.101	0.92	0.803	0.75	0.101	1.19	0.157	5.76	17.3	0	Complete
2 P-only DeepONet10	**long**	0.034	0.773	0.87	1.37	0.128	0.0762	0.058	0.0685	0.482	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.234	0.143	2.9	0.928	0.832	4.28	0.3	1.4	7.8	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0197	0.591	0.854	1.14	0.0875	0.122	0.0369	0.0377	0.426	0	Complete
4 T/P DeepONet10	**10k**	0.0871	0.306	1.15	0.406	0.106	0.998	0.107	2.11	2.76	0	Complete
4 T/P DeepONet10	**long**	0.0318	0.0936	0.327	0.0664	0.0608	0.439	0.0414	0.372	0.308	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	0.0113	0.0567	0.186	0.0697	0.0111	0.105	0.0136	0.111	0.106	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.0209	0.0376	0.204	0.0232	0.059	0.35	0.0283	0.233	0.213	0	Complete
6 T/P/ϕ DeepONet10	**10k**	0.024	0.107	0.479	0.124	0.0122	0.217	0.0275	0.295	0.232	0	Complete
6 T/P/ϕ DeepONet10	**long**	8.43e-03	0.0801	0.106	0.0407	0.0381	0.112	0.0126	0.0744	0.0497	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.0144	0.076	0.19	0.0787	0.012	0.145	0.0184	0.122	0.129	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	6.41e-03	0.0409	0.0796	0.0151	0.0166	0.0767	8.86e-03	0.033	0.0258	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.